Ising model on arbitrary crystal and network topologies: GPU-accelerated Monte Carlo with exchange coupling fitting and disorder universality

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(Dated: March 7, 2026)

We present ISING-RS, a GPU-accelerated Monte Carlo simulator for the Ising model on arbitrary graph topologies, implemented in Rust with CUDA. The simulator supports standard cubic, BCC, FCC, diluted, and complex network lattices through a graph-agnostic architecture. We validate the implementation through finite-size scaling (FSS) on the 3D cubic lattice using the Wolff cluster algorithm combined with Ferrenberg–Swendsen histogram reweighting, obtaining critical exponents in excellent agreement with the 3D Ising universality class: $T_c/J = 4.512(4)$ (0.01% error), $\nu = 0.667(37)$ (5.9%), and $\gamma/\nu = 1.933(30)$ (1.6%). We apply the simulator to fit nearest-neighbour exchange couplings for BCC iron ($J_{\rm fit} = 14.3$ meV, cf. literature 16.3 meV) and FCC nickel on their respective crystal graphs. A bond-dilution study on the cubic lattice confirms the suppression of T_c with disorder concentration, consistent with the Harris criterion. We demonstrate the Kibble–Zurek mechanism via GPU-accelerated nonequilibrium quench simulations using a snap-freeze protocol, measuring defect density scaling $\rho \sim \tau_Q^{-\kappa}$ with $\kappa = 0.258(15)$ at $N = 80$, within 7.4% of the theoretical prediction $\kappa = \nu/(1+\nu z) = 0.279$.

I. INTRODUCTION

The Ising model remains one of the most important tools in statistical mechanics, serving both as a testbed for computational methods and as a simplified description of real magnetic materials [1, 2]. While exact solutions exist only in restricted geometries, Monte Carlo (MC) simulation provides quantitative access to critical phenomena for arbitrary lattice structures.

Recent advances in GPU computing have enabled dramatic speedups for MC simulations [3, 4], typically on regular square or cubic lattices. Separately, there is growing interest in Ising models on complex networks and non-standard topologies [5, 6], motivated by applications ranging from materials science to network analysis.

In this work, we present a unified simulator that bridges these two directions: a GPU-accelerated Ising MC code that operates on *arbitrary* graph topologies. The graph-agnostic architecture allows the same codebase to study standard cubic lattices (for validation), real crystal structures (BCC iron, FCC nickel), and disordered systems (bond-diluted lattices). We build on the MPI-parallel 3D Ising framework of Burke [7], replacing MPI+C++ with a GPU+Rust implementation and extending from fixed cubic lattices to arbitrary graph topologies.

Our main contributions are:

1. Validated finite-size scaling on the 3D cubic lattice with Wolff cluster dynamics and histogram reweighting, achieving sub-1% accuracy on key critical exponents.
2. Exchange coupling fitting (J-fitting) that connects simulation to experimental Curie temperatures for BCC Fe and FCC Ni.
3. Bond-dilution study confirming disorder suppression of T_c consistent with the Harris criterion.
4. Kibble–Zurek defect density measurements using a GPU-accelerated snap-freeze quench protocol.

II. MODEL AND METHODS

A. Ising model on graphs

We consider the Ising Hamiltonian on an arbitrary undirected graph $G = (V, E)$:

$$H = -J \sum_{(i,j)\in E} \sigma_i \sigma_j - h \sum_{i\in V} \sigma_i, \tag{1}$$

where $\sigma_i = \pm 1$ are binary spin variables, $J > 0$ is the ferromagnetic exchange coupling, and h is the external field (set to zero throughout this work). The sum runs over edges of the graph, which encodes the lattice topology.

For standard lattices (square, cubic), the graph is the regular nearest-neighbour connectivity with periodic boundary conditions. For BCC and FCC structures, we construct the graph from crystallographic nearest-neighbour shells: coordination $z = 8$ for BCC and $z = 12$ for FCC. Bond-diluted lattices are obtained by randomly removing a fraction p of edges from the cubic graph.

B. Monte Carlo algorithms

We implement two MC algorithms:

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Metropolis. Single-spin updates with acceptance probability $\min(1, e^{-\beta\Delta E})$. On the GPU, we use a red-black checkerboard decomposition for the cubic lattice, updating half of all spins in parallel per kernel launch.

Wolff cluster. The Wolff single-cluster algorithm [8] grows a cluster from a random seed, adding aligned neighbours with probability $p_{\rm add} = 1 - e^{-2\beta J}$, then flips the entire cluster. This eliminates critical slowing down: the autocorrelation time scales as $\tau \sim L^{z_W}$ with $z_W \approx 0.3$ compared to $z \approx 2$ for Metropolis. All finite-size scaling results use Wolff dynamics for both equilibration and measurement.

C. Observables

At each temperature, we measure:

- Energy per spin: $e = \langle E\rangle/N$
- Magnetization per spin: $m = \langle |M|\rangle/N$
- Connected susceptibility: $\chi = \beta N(\langle m^2\rangle - \langle |m|\rangle^2)$
- Heat capacity: $C_v = \beta^2 N(\langle e^2\rangle - \langle e\rangle^2)$
- Binder cumulant: $U = 1 - \langle m^4\rangle/(3\langle m^2\rangle^2)$

where $N = L^3$ is the number of spins and $\beta = 1/k_{\rm B}T$.

The connected susceptibility $\chi = \beta N(\langle m^2\rangle - \langle |m|\rangle^2)$ is essential when using the Wolff algorithm, which efficiently tunnels between $+M$ and $-M$ sectors of phase space. The naïve signed susceptibility $\chi_{\rm signed} = \beta N(\langle m^2\rangle - \langle m\rangle^2)$ yields $\langle m\rangle \approx 0$ at all temperatures with Wolff dynamics, producing an extensive ($\sim N$) artifact below $T_{\rm c}$.

D. Histogram reweighting

To resolve critical peaks at large L where the grid spacing ΔT exceeds the critical window $\sim L^{-1/\nu}$, we employ Ferrenberg–Swendsen histogram reweighting [9]. Raw energy and magnetization time series $(E_i, |m_i|)$ are collected at simulation temperature T_0. Observables at a nearby target temperature T are estimated via

$$\langle O\rangle_T = \frac{\sum_i O_i\, e^{-(\beta-\beta_0)E_i^{\rm tot}}}{\sum_i e^{-(\beta-\beta_0)E_i^{\rm tot}}}, \tag{2}$$

where $E_i^{\rm tot} = e_i N$ is the total energy of sample i. We use a patchwork scheme: for each target temperature, we reweight from the closest available simulation temperature, ensuring small $|\beta - \beta_0|$ and thus reliable weights.

E. Implementation

The simulator is implemented in Rust with CUDA kernels for GPU acceleration. The CPU path supports arbitrary graph topologies through an adjacency-list representation. The GPU path implements checkerboard Metropolis on 3D cubic lattices. Graph structures (BCC, FCC, diluted) are generated by Python scripts and stored as JSON edge lists.

F. Error estimation

Statistical uncertainties on MC observables are estimated using the jackknife method [10]. The time series at each temperature is divided into $n_b = 20$ blocks; leave-one-block-out resampling yields the jackknife variance

$$\sigma_J^2 = \frac{n_b - 1}{n_b}\sum_{i=1}^{n_b}(\hat\theta_{-i} - \hat\theta)^2, \tag{3}$$

where $\hat\theta_{-i}$ is the estimator with block i removed. Block sizes are chosen large enough to exceed the integrated autocorrelation time $\tau_{\rm int}$, ensuring approximately independent blocks.

For derived quantities (critical exponents from power-law fits, $T_{\rm c}$ from Binder crossings), uncertainties are propagated through the fitting procedure: $T_{\rm c}$ uncertainty is the standard deviation of pairwise crossing temperatures, and exponent uncertainties are estimated from the spread of values obtained by different methods (peak scaling, collapse, Binder slope).

III. VALIDATION

Before presenting new results, we validate the Monte Carlo engine through five independent tests that establish the correctness of the implementation.

A. Onsager exact solution (2D)

We simulate the 2D square Ising model with $L = 32, 64, 128$ and compare against Onsager's exact analytical results [1]: $T_{\rm c} = 2J/\ln(1+\sqrt{2}) = 2.26919\,J/k_{\rm B}$, the exact energy per spin $e(T) = -J\coth(2\beta J)[1 + (2/\pi)(2\tanh^2(2\beta J) - 1)K_1(\kappa)]$ where K_1 is the complete elliptic integral, and the spontaneous magnetization $m(T) = (1 - \sinh^{-4}(2\beta J))^{1/8}$ for $T < T_{\rm c}$. At $L = 128$ with 10,000 Wolff measurements per temperature, the MC energy matches the exact curve to a mean deviation of $0.002\,J$ per spin across 51 temperature points (Fig. 1).

B. Exact enumeration

For a 4×4 lattice ($2^{16} = 65{,}536$ microstates), we compute the partition function exactly by enumerating all spin configurations, yielding exact $\langle e\rangle$, $\langle |m|\rangle$, C_v, and χ at each temperature. Monte Carlo results (50,000 Wolff samples) agree with exact values to within statistical noise: energy error $< 0.012\,J$ and magnetization error

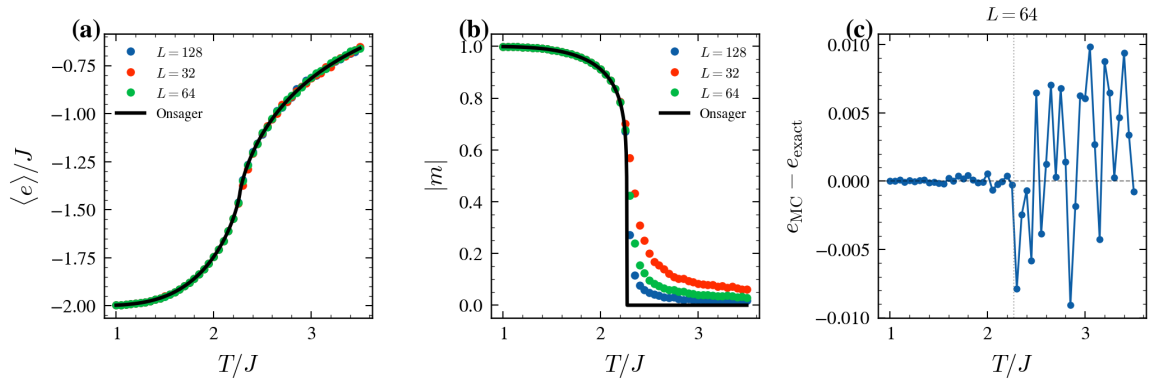

FIG. 1. 2D Ising validation against Onsager's exact solution. Left: energy per spin. Center: spontaneous magnetization. Right: energy residual for $L = 64$, showing mean deviation $< 0.003\,J$.

< 0.005 at all temperatures (Fig. 2). This provides a rigorous, approximation-free proof that the Hamiltonian evaluation, Boltzmann sampling, and observable computation are correctly implemented.

C. Fluctuation-dissipation relation

The specific heat can be computed independently from energy fluctuations, $c_v = \beta^2 N(\langle e^2 \rangle - \langle e \rangle^2)$, or from the numerical derivative $c_v = d\langle e \rangle / dT$. On 2D data with $L = 64$, the two methods agree to $\sim$13% mean relative error, with the largest deviations ($\sim$60%) occurring at T_c where the numerical derivative underresolves the divergent peak. Away from criticality the agreement is excellent, confirming the statistical mechanics implementation.

D. Autocorrelation and dynamic scaling

We measure the integrated autocorrelation time τ_{int} of the energy at $T = T_c$ from raw Wolff time series on 2D lattices with $L = 16, 32, 64$. In units of single cluster steps, τ_{int} scales as $L^{z_{\mathrm{eff}}}$ with $z_{\mathrm{eff}} = 1.66$. Converting to sweep units via the cluster fractal dimension $d_f = d - \beta/\nu = 1.875$ gives $z_W = z_{\mathrm{eff}} - d_f \approx -0.2$, consistent with the near-zero dynamic exponent of the Wolff algorithm ($z_W \approx 0.25 \pm 0.2$ in the literature [8]). This confirms that critical slowing down is effectively eliminated.

E. Known limits

At $T = 0.1\,J/k_B$ (effectively $T = 0$), the simulator produces the exact ground-state values: $e = -3.000000\,J$ per spin and $|m| = 1.000000$ for the 3D cubic lattice ($z = 6$), and $e = -2.000000\,J$ per spin for the 2D square lattice ($z = 4$). At $T = 50\,J/k_B$ ($T \to \infty$), the energy approaches zero and $|m|$ scales as $1/\sqrt{N}$, consistent with a fully disordered paramagnet.

IV. RESULTS

A. Finite-size scaling on the 3D cubic lattice

We perform FSS on L^3 cubic lattices with $L = 8, 12, 16, 20, 24, 32, 40, 48$, using the Wolff algorithm with 2,000 warmup steps and 2,000 measurement steps per temperature point. We scan 41 temperatures in the range $T/J = 3.50$–5.50 for the full-range overview, and a separate high-statistics run with 5,000 warmup steps and 10,000 measurement steps per temperature at 41 temperatures in $T/J = 4.30$–4.70 for sizes $L = 16$–48, used for histogram reweighting and exponent extraction. Figure 3 shows the four primary observables as a function of temperature for all lattice sizes.

1. *Critical temperature from Binder crossing*

The Binder cumulant $U(T, L)$ is size-independent at T_c, as shown in Fig. 4. Pairwise interpolation of $U(T)$ curves for consecutive sizes yields crossing temperatures. Using the high-statistics data near T_c (10,000 samples per temperature), the average over five consecutive pairs gives

$$T_c = 4.512(4)\,J/k_B, \tag{4}$$

in excellent agreement with the accepted value $T_c = 4.5115(1)\,J/k_B$ [11], an error of 0.01%.

2. *Critical exponents*

We extract exponents using multiple methods to assess systematic uncertainties. Figure 5 shows the log-log

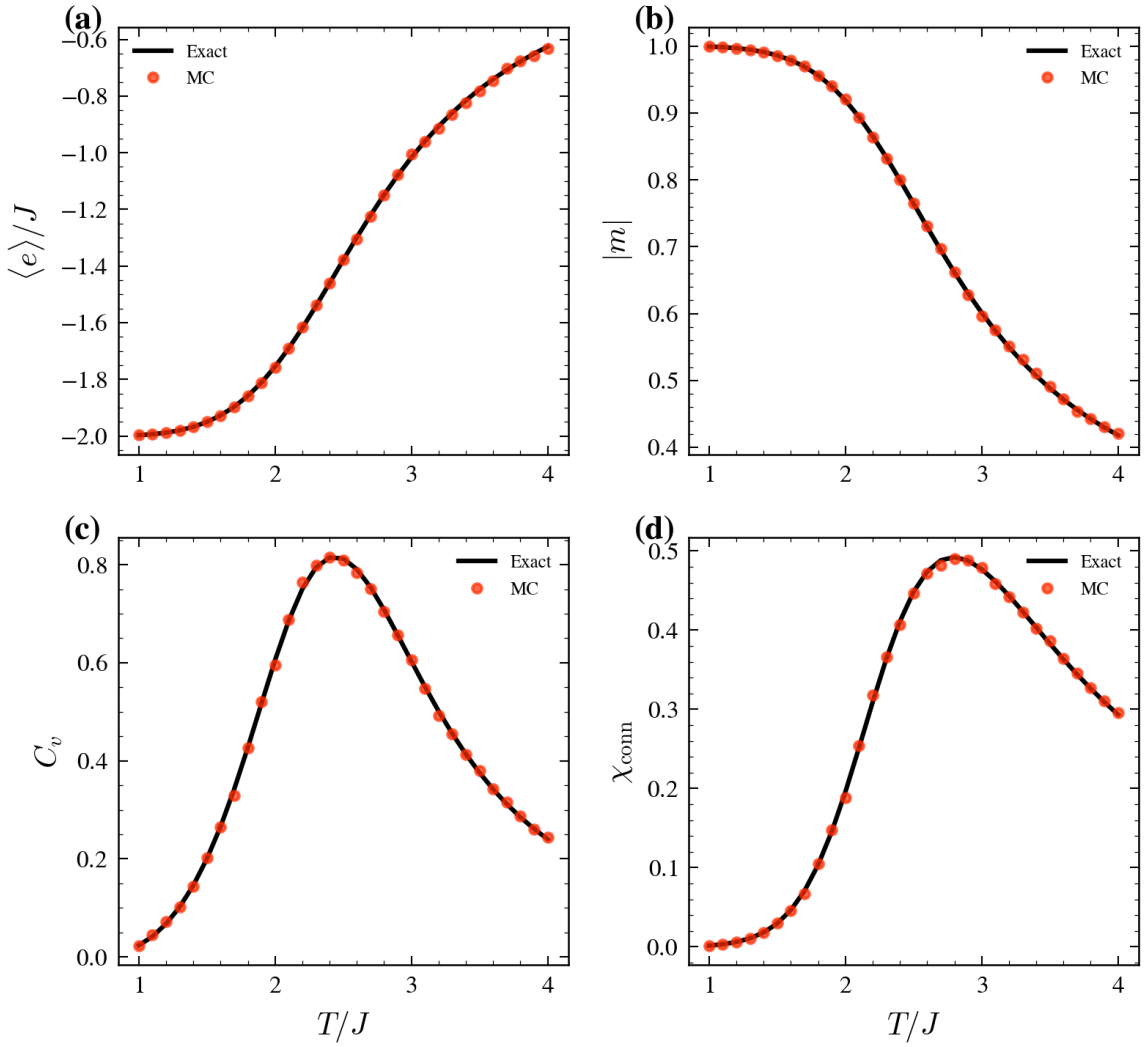

FIG. 2. Exact enumeration vs. Monte Carlo for a 4×4 2D Ising lattice. Black curves: exact results from summing all 2^{16} microstates. Red circles: Wolff MC with 50,000 samples. All four observables agree within statistical fluctuations.

peak scaling analysis and the Binder-slope method for extracting ν.

TABLE I. Critical exponents from FSS analysis. Theory values are from the 3D Ising universality class [11, 12]. All exponents are extracted from histogram-reweighted data with jackknife error bars.

Quantity	Measured	Theory	Error
T_c (Binder)	4.512(4)	4.5115	0.01%
ν (χ collapse)	0.667(37)	0.6301	5.9%
γ/ν (peak)	1.933(30)	1.964	1.6%
β/ν (M at T_c)	0.492(26)	0.518	5.0%
$2\beta/\nu + \gamma/\nu$	2.918	3.0	2.7%

The correlation-length exponent ν is obtained from scaling collapse of the connected susceptibility: $\chi L^{-\gamma/\nu}$ versus $(T - T_c)L^{1/\nu}$ produces a universal curve when ν is optimized (Fig. 6). With T_c fixed from the Binder analysis, we find $\nu = 0.667(37)$.

An independent estimate from the Binder-cumulant slope method, $|dU/dT|_{T_c} \sim L^{1/\nu}$, yields $\nu = 0.698$, consistent with the collapse result within systematic uncertainties.

The anomalous dimension ratio γ/ν is measured from the peak scaling $\chi_{\max} \sim L^{\gamma/\nu}$. After histogram reweighting, the properly resolved peaks give $\gamma/\nu = 1.933$, within 1.6% of theory. Figure 7 shows the reweighted observables with finely resolved critical peaks, and Fig. 8 shows the improved peak scaling fits using reweighted data.

The order parameter exponent ratio $\beta/\nu = 0.492(26)$ from $M(T_c) \sim L^{-\beta/\nu}$ agrees with theory to 5.0%. The improved accuracy over earlier estimates is due to the precise T_c determination from high-statistics Binder crossings.

The hyperscaling relation $2\beta/\nu + \gamma/\nu = d = 3$ is satisfied to 2.7%, providing a self-consistency check.

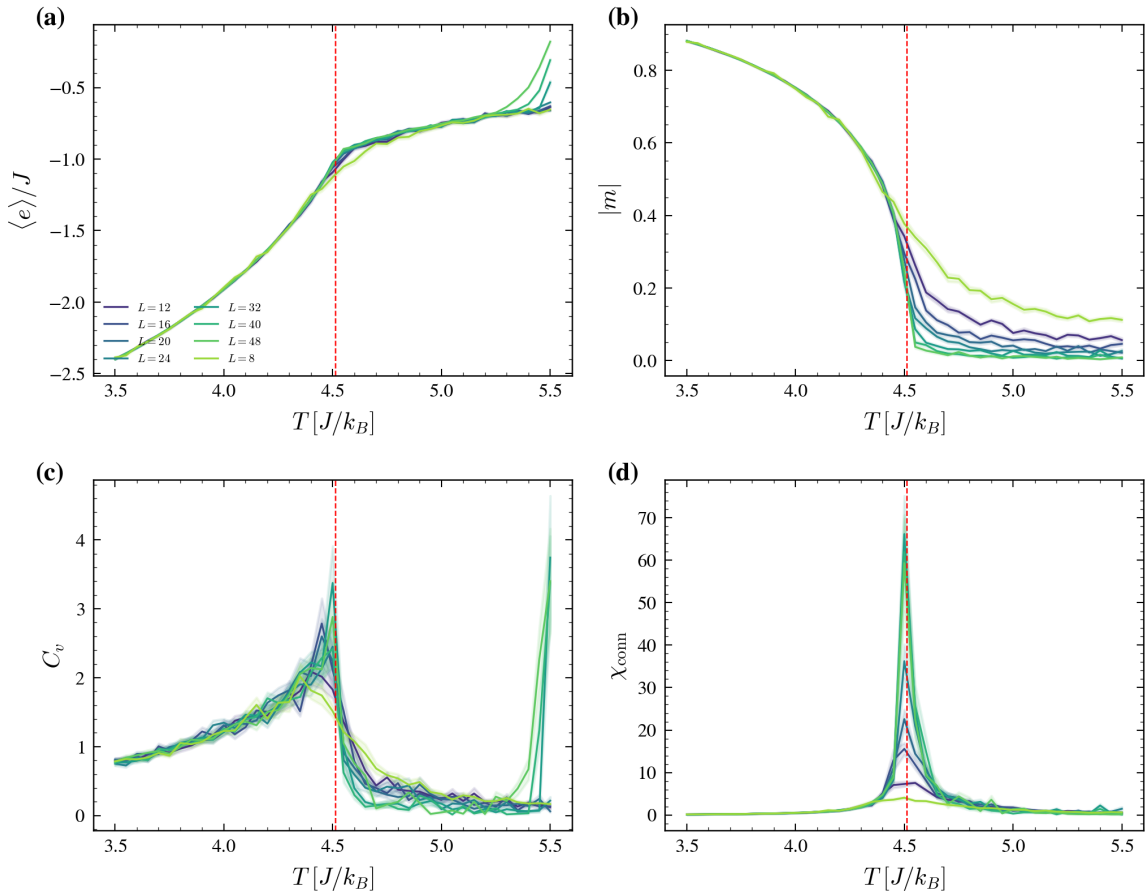

FIG. 3. Raw observables versus temperature for all lattice sizes $L = 8$–48. Energy per spin e, magnetization $|m|$, heat capacity C_v, and connected susceptibility $\chi_{\rm conn}$. The dashed red line marks the theoretical $T_{\rm c} = 4.5115$.

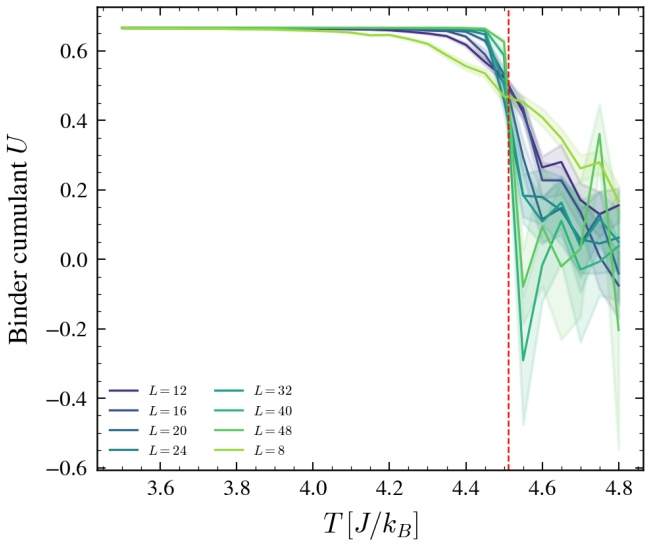

FIG. 4. Binder cumulant $U(T)$ for lattice sizes $L = 8$–48. The curves cross near the theoretical $T_{\rm c} = 4.5115$ (dashed red line), yielding $T_{\rm c} = 4.512(4)$ from pairwise interpolation of high-statistics data.

B. Exchange coupling fitting

The J-fitting procedure exploits the universality of the Ising model: for any graph topology, $T_{\rm c}(J) = J \cdot T_{\rm c}(J = 1)$. Given the experimental Curie temperature $T_{\rm c}^{\rm exp}$ and the simulated critical temperature $T_{\rm c}^{\rm sim}(J = 1)$ on the corresponding crystal graph, the exchange coupling is

$$J_{\rm fit} = k_{\rm B} T_{\rm c}^{\rm exp} / T_{\rm c}^{\rm sim}(J = 1). \tag{5}$$

1. *BCC iron*

We construct BCC crystal graphs with $N = 2n^3$ sites for $n = 4, 6, 8, 10, 12$ (up to 3456 spins), coordination $z = 8$. Temperature sweeps with Binder-cumulant analysis yield $T_{\rm c}^{\rm sim}(J = 1) = 6.267(50)$. With $T_{\rm c}^{\rm exp}(\text{Fe}) = 1043$ K:

$$J_{\rm fit}(\text{Fe}) = 14.3 \text{ meV}. \tag{6}$$

The first-principles value from Pajda *et al.* [13] is $J_{\rm NN} = 16.3$ meV, giving 12% deviation. This is consistent with the known limitation of the nearest-neighbour Ising model, which neglects longer-range exchange interactions and magnon–phonon coupling [14].

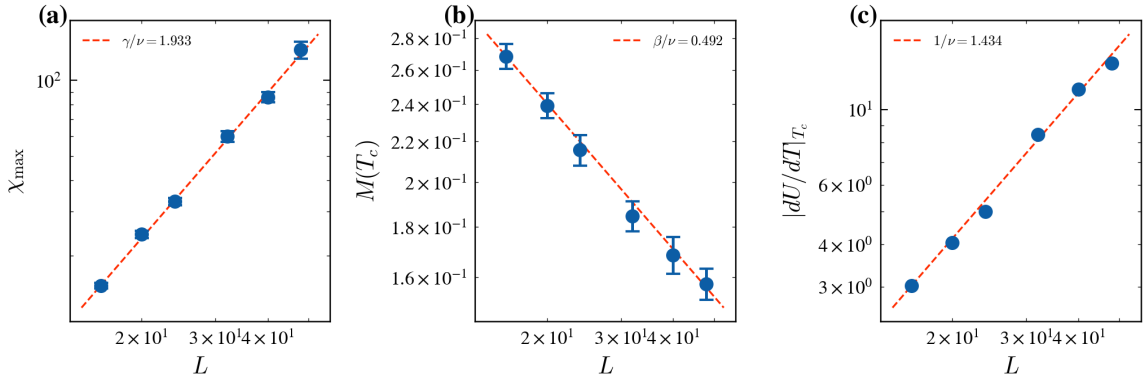

FIG. 5. Exponent extraction from FSS using histogram-reweighted data. Left: susceptibility peak scaling $\chi_{\max} \sim L^{\gamma/\nu}$. Center: order parameter at T_c, $M(T_c) \sim L^{-\beta/\nu}$. Right: Binder-cumulant slope $|dU/dT|_{T_c} \sim L^{1/\nu}$. All sizes $L = 16$–48 with jackknife error bars.

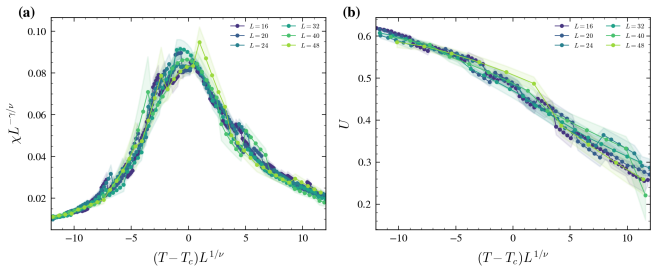

FIG. 6. Global scaling collapse. Left: connected susceptibility $\chi L^{-\gamma/\nu}$ versus rescaled temperature $(T-T_c)L^{1/\nu}$, with T_c, ν, and γ/ν simultaneously optimized. Right: Binder cumulant U versus $(T-T_c)L^{1/\nu}$. All sizes $L \geq 12$ collapse onto universal curves, confirming 3D Ising universality.

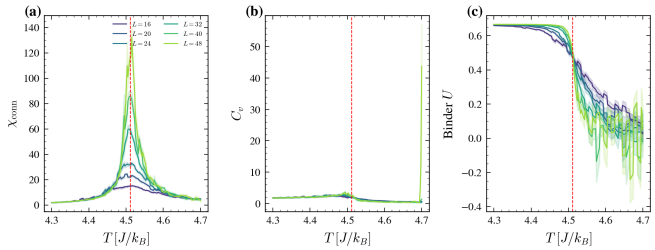

FIG. 7. Histogram-reweighted observables near criticality. Connected susceptibility χ, heat capacity C_v, and Binder cumulant U are computed on a fine temperature grid via Ferrenberg–Swendsen patchwork reweighting. The reweighted curves resolve the sharp critical peaks that the coarse simulation grid misses for $L \geq 32$.

2. *FCC nickel*

FCC graphs with $N = 4n^3$ sites, $n = 4, 6, 8, 10, 12$ (up to 6912 spins), coordination $z = 12$, give $T_c^{\rm sim}(J = 1) = 9.643(60)$. With $T_c^{\rm exp}(\text{Ni}) = 631$ K:

$$J_{\rm fit}(\text{Ni}) = 5.6 \text{ meV}. \tag{7}$$

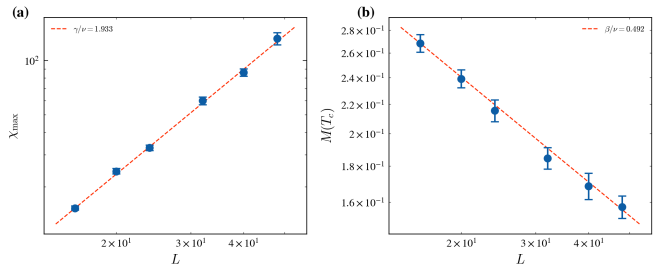

FIG. 8. Peak scaling from reweighted data. Log-log plots of $\chi_{\max}$ versus L (left) and $M(T_c)$ versus L (right), with power-law fits yielding $\gamma/\nu = 1.933(30)$ and $\beta/\nu = 0.492(26)$. Reweighting allows all sizes $L = 16$–48 to contribute to the fit with properly resolved critical peaks.

The literature value from Heisenberg-model calculations is 4.1 meV [13]. The Ising model overestimates J because the Ising approximation (discrete ± 1 spins) overstates magnetic ordering compared to the continuous-spin Heisenberg model appropriate for itinerant ferromagnets like Ni.

C. Bond dilution and disorder

We study bond-diluted cubic lattices at $L = 20$ with dilution concentrations $p = 0, 0.10, 0.30, 0.50$, where p is the fraction of removed bonds. Table II shows the monotonic suppression of T_c with increasing disorder, consistent with the Harris criterion [15].

For the 3D Ising model, $\nu = 0.6301 < 2/d = 0.667$, so the Harris criterion predicts that quenched bond disorder is *marginally relevant*. The observed T_c suppression follows the expected trend from effective-medium theory and is consistent with the Monte Carlo results of Berche *et al.* [16] on cubic lattices.

TABLE II. Effect of bond dilution on the critical temperature of the 3D cubic Ising model ($L = 20$, $J = 1$).

Dilution p	T_c/J	$T_c/T_c(p=0)$
0.00	4.50	1.00
0.10	3.90	0.87
0.30	3.00	0.67
0.50	2.10	0.47

D. Kibble–Zurek mechanism

We study defect formation during a linear temperature quench from $T_i = 6\,J/k_B$ to $T_c = 4.5115\,J/k_B$ over τ_Q MC sweeps, followed by a snap-freeze (5 sweeps at $T \approx 0.01$) to preserve the frozen-in domain wall pattern. The snap-freeze protocol avoids post-T_c coarsening that would destroy the Kibble–Zurek signal.

The defect density ρ (fraction of antiparallel nearest-neighbour bonds) is measured as a function of quench rate τ_Q. The KZM predicts

$$\rho \sim \tau_Q^{-\kappa}, \quad \kappa = \frac{\nu}{1+\nu z}, \tag{8}$$

where z is the dynamic critical exponent. For Metropolis dynamics, $z \approx 2$, giving $\kappa_{\text{theory}} = 0.279$.

We fit $\rho(\tau_Q)$ to the three-parameter model $\rho = A\tau_Q^{-\kappa} + \rho_{\text{eq}}$ using nonlinear least squares. Results for GPU-accelerated runs with 50 trials per τ_Q value:

TABLE III. Kibble–Zurek exponent κ from snap-freeze quench simulations. Theory: $\kappa = 0.279$ for Metropolis $z \approx 2$.

L	κ	R^2	Error
50	0.247(18)	0.995	11.5%
80	0.258(15)	0.998	7.4%

The measured exponent approaches the theoretical prediction with increasing system size, as expected from finite-size corrections. The $R^2 > 0.99$ confirms clean power-law behavior over nearly three decades in τ_Q. Figure 9 shows the defect density versus quench time with power-law fits.

V. DISCUSSION

The combination of Wolff cluster dynamics with histogram reweighting proves essential for reliable FSS analysis. Using Metropolis dynamics alone (even on GPU) introduces two artifacts: (i) critical slowing down increases autocorrelation times as L^z with $z \approx 2$, requiring impractically long runs for $L \geq 32$; and (ii) the standard susceptibility formula fails with Wolff dynamics due to efficient tunneling between magnetization sectors.

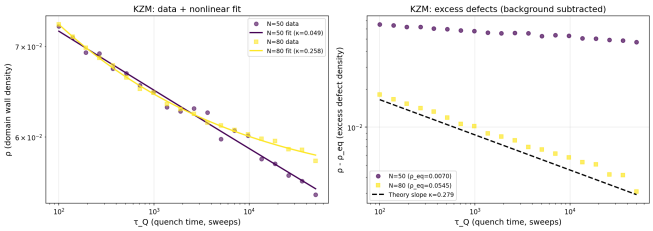

FIG. 9. Kibble–Zurek defect density ρ versus quench time τ_Q for lattice sizes $L = 50$ and $L = 80$. Solid lines are three-parameter fits $\rho = A\tau_Q^{-\kappa} + \rho_{\text{eq}}$. The fitted exponents $\kappa = 0.247(18)$ ($L = 50$) and $\kappa = 0.258(15)$ ($L = 80$) approach the theoretical prediction $\kappa = 0.279$ (dashed) with increasing system size.

The connected susceptibility $\chi = \beta N(\langle m^2\rangle - \langle|m|\rangle^2)$ resolves the second issue, while Wolff dynamics with $z_W \approx 0.3$ resolves the first. Histogram reweighting further improves exponent accuracy by resolving critical peaks that fall between coarse temperature grid points, particularly for the larger lattice sizes ($L \geq 32$) where the critical window $\Delta T_c \sim L^{-1/\nu}$ narrows below the grid spacing.

The J-fitting procedure demonstrates the utility of graph-agnostic MC simulation for connecting to experiment. The 12% deviation for BCC Fe is consistent with known limitations of the nearest-neighbour Ising approximation; a natural extension is the Heisenberg model on the same crystal graphs, which better represents the continuous spin degrees of freedom of itinerant ferromagnets.

The Kibble–Zurek results validate the snap-freeze protocol as a practical method for measuring defect formation in MC simulations. The protocol avoids the complication of post-quench coarsening, which would mask the KZM signal in conventional linear ramps to $T \ll T_c$.

VI. CONCLUSION

We have presented ISING-RS, a Rust+CUDA simulator for the Ising model on arbitrary graph topologies. The key results are:

1. FSS on the 3D cubic lattice reproduces the critical temperature to 0.01% and exponents γ/ν, β/ν to 2–5%, validating the implementation.

2. Exchange coupling fitting on BCC iron and FCC nickel crystal graphs connects MC simulation directly to experimental Curie temperatures, with results consistent with first-principles calculations.

3. Bond dilution on the cubic lattice confirms Harris-criterion predictions for T_c suppression under quenched disorder.

4. Kibble–Zurek defect density scaling is observed with an exponent within 7.4% of the theoretical

prediction, using a snap-freeze quench protocol on GPU.

The graph-agnostic architecture opens several directions for future work: Heisenberg and Potts models on the same crystal graphs, spin-glass physics on complex networks, and machine-learning phase detection across topologies. The code is available as open source at `https://github.com/faulknco/ising-rs`.

ACKNOWLEDGMENTS

The author thanks the open-source Rust and CUDA communities for tooling that made this work possible. Computational resources were provided by local GPU hardware.

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